Extended Propagation Risk Modeling of Invasive Spotted Lanternflies (*Lycorma delicatula*) Using Real-Time Observation Data

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Abstract

Lycorma delicatula is an insect invasive to the United States that is causes millions of dollars in annual losses to individual states' agricultural industries(1; 2; 3; 4; 5). It parasitizes various host plants used in food, timber, and ornamental plant production(6; 5). Feeding may stunt the growth and yield of host plants and provides an environment for sooty mold to grow, which can reduce photosynthesis(7; 8). The continuing invasion requires further monitoring because many treatment methods are being studied but not yet deployed. Because such methods are expensive or not yet approved for use on crops or wild areas(9; 10; 11; 12; 13; 14; 15), an interactive website that shows areas of high risk for infestation allows agricultural managers to make informed decisions on when, where, and how to treat their crops. Websites are effective for allowing citizens to report verifiable sightings which can be used in expanded propagation risk models. Prior work developing machine learning models predicting local *L. delicatula* propagation risk levels(5; 16) will expand with recent data and from submissions through a reporting form on the website. This reporting method can simultaneously update the training data of propagation risk models made from 2014–2025 sightings to make them more accurate while notifying relevant authorities of invasions in their area.

Background

The spotted lanternfly *Lycorma delicatula* is a planthopper that was introduced to Berks County, Pennsylvania in 2014 and has since invaded many states(1; 2; 5). They are now found in the East Coast and Midwest regions of the United States and through the southern Canadian border, feeding on at least 50 species of host plants in the country(1; 2; 5). Many host plants are native to the United States or have agricultural significance as sources of fruit, vegetables, medicine, timber, paper, fiber, and landscape beauty(6). *L. delicatula* inserts its proboscis through the outer layer of a leaf, branch or trunk into the phloem layer, reducing the amount of nutrient supply flowing through the plant(2). If enough lanternflies feed on a plant for a prolonged period of time, they can negatively impact the yield from the plant and stunt its overall growth(17). In addition, *L. delicatula* secrete sugar-rich honeydew from their abdomens as a waste product, and this which accumulates on lower parts of the plants, facilitating the growth of sooty mold(2); this mold can cover leaves and impede photosynthesis, which are necessary metabolic processes in plants(7; 8).

Management of *L. delicatula* is a difficult problem due to multiple factors, namely the urban adaptations of the United States and Canada population as seen in phenological analysis(18). Strategies implemented against other invasive species are still in consideration and testing phases, including the use of entomopathogenic fungi(9; 10; 11; 12), parasitic wasps(13; 14; 15), insecticides(19; 20; 21; 22), traps(23; 24; 25; 26), and more. Host plant removal may be expensive for large swaths of wild areas and impact species richness; for cultivated plants, this may also be unaffordable for farms unless incentivized. On top of the potential treatments still being constrained, *L. delicatula* can use human-mediated transportation to travel long distances; they can lay eggs on or cling to modes of transit like cars, trucks, trains, boats, and their cargo while the vehicle travels up to 100 kilometers per hour(27; 28).

Machine learning is applicable to modeling the propagation behavior of species, and several techniques are capable of handling spatiotemporal data and time series. Because *L. delicatula* tend to produce only one generation per year, risk classifications can represent states in a Markov chain where each step is represented in an individual year, given anomalous hatching times do not have a noticeable effect on the amount of generations per year(29; 1; 2). Markov chain representations of real-world phenomena are simple to implement in classification-based machine learning models, such as artificial neural networks (ANNs)(30). While long short-term memory (LSTM) modeling techniques are often used in similar situations because the models constructed from them depend on time series data(31), multi-layer perceptrons (MLPs) and random forest (RF) decision tree models were implemented in previous works(5). MLPs and the RF decision trees allowed individual, smaller models to be ensembled to reduce the bias that would otherwise be seen in models that are not part of an ensemble(32; 33). These models can be modified to include aggregated data from future years to extend their predictive capacity up to the year after the final year in which training data is added, so these models can be updated with real-time observation data.

Significance

Damage due to *L. delicatula* infestations reached an estimated US$324 million in losses to Pennsylvanian industries by 2019(3; 4). Additionally, in the first three respective years of infestation in the Finger Lakes region, New York State Integrated Pest Management reported losses to grape production as US$1.5 million, US$4 million, and US$8.8 million(17; 34). The aforementioned combination of factors complicating invasion mitigation makes it necessary to continue modeling the infestation risk levels of certain locations.

Building on the work completed in 2025, where the new machine learning models were the first of their kind in terms of the combination of technique (multi-layer perceptrons and random forest decision tree models) and application (propagation risk level prediction for *L. delicatula*), the work for this project will be one of the first to create a sighting reporting system that can directly inform the stakeholders that are most affected by the invasion.

No websites or online resources currently have the capability of showing rich data on the species and its invasion in the contiguous United States to the degree proposed here. The website, if marketed effectively, should close the gap between the information that people collect on *L. delicatula*, such as those by citizen scientists and researchers, and what gardeners and agriculture-related business owners need to produce an optimal yield.

Additionally, the amount of publicly available data on *L. delicatula* sightings may not be sufficient for accurate predictions, making it necessary to have a dedicated reporting form to supplement data extracted from other sources like iNaturalist and EDDMapS(35; 36).

Project Design

During the summer 2025 Provost URI program, I developed machine learning models using random forest decision trees and multi-layer perceptrons to classify invasion propagation risk levels in states and counties for the year of 2025. Expansions to the machine learning models made in prior research will include, at minimum:

(1) adding training data manipulated from raw data by the same sources used for the models developed during the summer 2025 URI program, including 2025 sightings.

(2) comparing 2025's resulting estimated population densities and their associated propagation risk levels with that of the predictions made for that year using data only from 2014–2024.

A desktop-accessible website meeting the following minimum specifications will be publicly available alongside the abstract and presentation required for the program:

(1) a website hosted under a new, purchased domain through a reputable registrar (likely Cloudflare) with functionality to prevent denial of service incidents and to allow for easy access to website usage analytics to track the amount of sighting reports made using the website.

(2) a "home" page showing a map of the contiguous United States with interactive features, such as toggling map layers. Major options include the estimated population densities (calculated in Python from the 2014–2025 training data) and the propagation risk level outputs from the older machine learning models.

(3) clickable functionality that queries the exact risk levels and population densities of a specific area of the contiguous United States for each year of the invasion from the machine learning model training data.

(4) resource pages to provide specific information on *L. delicatula* reporting methods to users from a specified area, providing separate resources for regular civilians and gardeners who can be impacted by the pest.

(5) resource pages to provide specific information on reporting and treatment options available to farms and their managers, informing them of their expected efficacy as seen in prior literature.

(6) a reporting form that allows users to provide all relevant information about a sighting and directs the user to inform the relevant invasion monitoring services, linking to additional forms.

(7) a support ticket system that lets users request resource corrections, ask for help reporting their sightings, etc.

Data management plan: A data management plan is already available for the training data that is in use with the current machine learning models. All data comes from publicly available sources or sources that were once publicly available, such as removed USDA databases and is manipulated into condensed training data that excludes any individual or group's personal information. Raw (non-manipulated data) contains only anonymized personal information from observers of *L. delicatula* and is only available upon request. No data collected from new sources will be made available without making personal information anonymized. New training data will be made available on a GitHub repository alongside the new models' Jupyter Notebooks, website code, and relevant documentation until at least July 2028. All dependencies and resources used to build the website and predictive models will be cited.

Animal handling: No handling of animals is needed because all tasks for this project can be done without direct interaction with *L. delicatula*. Individuals who report sightings of *L. delicatula* through the website are not participating in the tasks of this research, can do so without handling the insect, and are not likely to come to harm in any way by handling the insect.

Expected Outcomes and Deliverable

This research builds on funding and support from the summer 2025 Provost URI program, which involved the creation of iteratively improved machine learning models predicting the spread of *L. delicatula* across the contiguous United States. They will be hosted on the same repository.

The improved machine learning models, originally developed in previous research, will expand due to report form responses and other sources; these updated models may be used to predict the propagation risk levels for 2027, which is something the previously available data was not able to do because predictions can only be made one generation in advance.

The other primary deliverable for this project is to be located in a website meeting the minimum specifications listed above. The resource pages will contain summaries of literature that are most relevant to U.S. agricultural managers (sorted by industry) and regular citizens based on their role to play in managing the *L. delicatula* invasion; extracted analytics on website usage will guide improvements to the website in terms of ease of use and how helpful/informational it is to its intended audiences. Analytics summaries will be provided at the end of the research period.

The reporting form would in turn expand a curated dataset of observations of *L. delicatula* available through iNaturalist alongside an existing iNaturalist project using a form that anyone can fill when they encounter a spotted lanternfly anywhere in the contiguous United States.

The data flowing from the user is as follows: users supply sighting data in the reporting form and to other reporting agencies linked within the website. The information sent to the built-in reporting form is stored on a server (storage has been procured), and I repeatedly convert that data into training data for revised models. The revised models are mapped and downloaded, then manually re-uploaded to the website to update the maps.

Observations made through this form may be sent to iNaturalist which in turn sends data to the GBIF, which both benefit biodiversity and conservation researchers by providing observation data under a "free" license.

The underlying code behind the website and services created for this research will be hosted on the aforementioned GitHub repository(37) under the MIT License(38), so they can be audited.

References

[1] L. E. Barringer, L. Donovall, S.-E. Spichiger, D. Lynch, and D. Henry, "The first new world record of lycorma delicatula (insecta: Hemiptera: Fulgoridae)," *Entomological News*, vol. 125, no. 1, pp. 20–23, 2015, accessed: 2025-11-19. [Online]. Available: https://www.researchgate.net/publication/279531772_The_First_New_World_Record_of_Lycorma_delicatula_Insecta_Hemiptera_Fulgoridae

[2] S. K. Dara, L. Barringer, and S. P. Arthurs, "Lycorma delicatula (hemiptera: Fulgoridae): A new invasive pest in the united states," *Journal of Integrated Pest Management*, vol. 6, no. 1, p. 20, 2015, accessed: 2025-05-20. [Online]. Available: https://flow.hemiptera-databases.org/flowpdf/3413.pdf

[3] J. K. Karper, W. Stone, T. W. Kelsey, and L. F. Kime, "Potential economic impact of the spotted lanternfly on agriculture and forestry in pennsylvania," Dec 2019, accessed: 2025-12-02. [Online]. Available: https://www.rural.pa.gov/download.cfm?file=Resources/reports/assets/25/Spotted-Lanternfly-2019.pdf

[4] D. Borowy, "Spotted lanternfly in perspective," Feb 2021, accessed: 2025-12-02. [Online]. Available: https://www.nps.gov/articles/000/slf-in-perpective.htm

[5] E. Keklak and W. S. Cai, "Predictive model of the spread of the spotted lanternfly in the continental united states using machine learning," Apr 2025.

[6] L. Barringer and C. M. Ciafré, "Worldwide feeding host plants of spotted lanternfly, with significant additions from north america," *Environmental Entomology*, vol. 49, no. 5, p. 999–1011, Aug 2020, accessed: 2025-05-20. [Online]. Available: https://doi.org/10.1093%2Fee%2Fnvaa093

[7] I. Team, "Are spotted lanternflies dangerous to the environment?" 2025, accessed: 2026-02-27. [Online]. Available: https://iere.org/are-spotted-lanternflies-dangerous-to-the-environment/#FAQ_10_What_is_sooty_mold_and_why_is_it_a_problem

[8] "Spotted lanternfly damage," accessed: 2026-02-27. [Online]. Available: https://cals.cornell.edu/integrated-pest-management/outreach-education/whats-bugging-you/spotted-lanternfly/spotted-lanternfly-damage

[9] A. G. Eric H. Clifton, Louela A. Castrillo and A. E. Hajek, "A pair of native fungal pathogens drives decline of a new invasive herbivore," *Applied Biological Sciences*, vol. 116, no. 19, pp. 9178–9180, Apr 2019, accessed: 2026-01-20. [Online]. Available: https://www.pnas.org/doi/full/10.1073/pnas.1903579116

[10] K. Ramanujan, "Destructive plant pest thwarted by two native fungi," Apr 2019, accessed: 2026-01-20. [Online]. Available: https://news.cornell.edu/stories/2019/04/destructive-plant-pest-thwarted-two-native-fungi

[11] N. E. J. R. T. R. J. P. R. Eric H Clifton, Ann E Hajek and D. J. Biddinger, “Applications of beauveria bassiana (hypocreales: Cordycipitaceae) to control populations of spotted lanternfly (hemiptera: Fulgoridae), in semi-natural landscapes and on grapevines,” *Environmental Entmology*, vol. 49, no. 4, p. 854–864, Jun 2020, accessed: 2026-01-20. [Online]. Available: https://academic.oup.com/ee/article/49/4/854/5850580

[12] A. E. Hajek, T. A. Everest, and E. H. Clifton, “Accumulation of fungal pathogens infecting the invasive spotted lanternfly, lycorma delicatula,” *Insects*, vol. 14, no. 12, Nov 2023, accessed: 2026-01-20. [Online]. Available: https://www.mdpi.com/2075-4450/14/12/912

[13] F. G. Marco and M. Hoddle, “Proactive biological control of spotted lanternfly: parasitism and host feeding behavior of anastatus orientalis (hymenoptera: Eupelmidae) on lycorma delicatula (hemiptera: Fulgoridae) egg masses,” *Biological Control*, vol. 195, p. 105551, 2024, accessed: 2025-05-20. [Online]. Available: https://doi.org/10.1016/j.biocontrol.2024.105551

[14] M. M. Mari West and M. S. Hoddle, “Proactive assessment of a native north american egg parasitoid, anastatus reduvii (hymenoptera: Eupelmidae), as a potential biological control agent of lycorma delicatula (hemiptera: Fulgoridae), in california,” vol. 200, p. 105687, Jan 2025, accessed: 2026-01-21. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1049964424002524

[15] M. Molfini, M. West, F. Gómez-Marco, A. Iacovone, and M. S. Hoddle, “Proactive evaluation of a native european parasitoid, anastatus bifasciatus (hymenoptera: Eupelmidae), for biological control of lycorma delicatula (hemiptera: Fulgoridae),” *Biological Control*, vol. 203, p. 105730, Feb 2025, accessed: 2026-02-04. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1049964425000404

[16] *2025 UNDERGRADUATE SUMMER RESEARCH AND INNOVATION SYMPOSIUM*, New Jersey Institute of Technology. New Jersey Institute of Technology, Jul 2025. [Online]. Available: https://research.njit.edu/uri/sites/research.uri/files/2025%20BOOK%20OF%20ABSTRACTS.pdf

[17] A. F. Pinto, B. C. Eshenaur, F. E. Acevedo, A. A. Calixto, M. Centinari, and M. I. Gómez, “Assessing the potential economic impacts of spotted lanternfly (hemiptera: Fulgoridae) infestations on grape production in new york state,” *Journal of Integrated Pest Management*, vol. 16, no. 1, p. 2, Jan 2025, accessed: 2025-12-02. [Online]. Available: https://academic.oup.com/jipm/article/16/1/2/7964417?searchresult=1&login=false

[18] H. L. Owen, F. Meng, and K. M. Winchell, “Urbanization and environmental variation drive phenological changes in the spotted lanternfly, lycorma delicatula (hemiptera: Fulgoridae),” *Biological Journal of the Linnean Society*, vol. 143, no. 4, Dec 2024, accessed: 2026-02-27. [Online]. Available: https://academic.oup.com/biolinnean/article/143/4/blae099/7908807

[19] S.-G. L. S.-C. S. J. K. JI-Doo Park, Min-Young Kim and I.-K. Park, “Biological characteristics of lycorma delicatula and the control effects of some insecticides,” *Korean Journal of Applied Entomology*, vol. 48, no. 1, pp. 53–57, 2009, accessed: 2026-01-06. [Online]. Available: https://d1wqtxts1xzle7.cloudfront.net/76195709/Park.2009.KoreanAppEnt.

SLF_.BiologyInsecticides.TRANS_-libre.pdf?1639368011=&response-content-disposition=inline%3B+filename%3DLycorma_delicatula.pdf&Expires=1767756917&Signature=aL1SK6qGbaDJ7p7z6AuzyFzG-dyhKOiL9Q23vS~8cPiUorTYTcf752cfjLvYpGkHR4UJAUskS2wxEuLfz_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

[20] J.-W. J. Y.-H. S. S.-R. C. K.-S. A. J.-O. Y. Changmann Yoon, Sang-Rae Moon and G.-H. Kim, "Repellency of lavender oil and linalool against spot clothing wax cicada, lycorma delicatula (hemiptera: Fulgoridae) and their electrophysiological responses," *Journal of Asia-Pacific Entomology*, vol. 14, no. 4, pp. 411–416, December 2011, accessed: 2025-11-19. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S1226861511000781

[21] H. Leach, D. J. Biddinger, G. Krawczyk, E. Smyers, and J. M. Urban, "Evaluation of insecticides for control of the spotted lanternfly, lycorma delicatula, (hemiptera: Fulgoridae), a new pest of fruit in the northeastern us," *Crop Protection*, vol. 124, p. 104833, 2019, accessed: 2025-05-20. [Online]. Available: https://doi.org/10.1016/j.cropro.2019.05.027

[22] L. J. Nixon and T. C. Leskey, "Evaluation of insecticide residues against spotted lanternfly (hemiptera: Fulgoridae)," *Journal of Economic Entomology*, vol. 117, no. 4, pp. 1582–1587, May 2024, accessed: 2025-11-19. [Online]. Available: https://doi.org/10.1093/jee/toae106

[23] L. J. Nixon, C. Barnes, E. Deecher, K. Madalinska, A. Nielsen, J. Urban, and T. C. Leskey, "Evaluating deployment strategies for spotted lanternfly (lycorma delicatula hemiptera: Fulgoridae) traps," *Journal of Economic Entomology*, vol. 116, no. 2, pp. 426–434, Mar 2023, accessed: 2026-02-27. [Online]. Available: https://bioone.org/journals/journal-of-economic-entomology/volume-116/issue-2/toad038/Evaluating-deployment-strategies-for-spotted-lanternfly-Lycorma-delicatula-Hemiptera/10.1093/jee/toad038.short

[24] E. Swackhamer, A. Korman, H. L. Leach, and J. A. Francese, "How to build a spotted lanternfly circle trap," Feb 2024, accessed: 2026-02-27. [Online]. Available: https://extension.psu.edu/how-to-build-a-spotted-lanternfly-circle-trap

[25] M. F. Cooperband and K. M. Murman, "Improving traps for spotted lanternflies, lycorma delicatula (hemiptera: Fulgoridae), by leveraging their own signals," *Insects*, vol. 16, no. 9, p. 930, Sep 2025, accessed: 2026-02-27. [Online]. Available: https://www.mdpi.com/2075-4450/16/9/930

[26] K. Maxwell, "Building effective traps to control spotted lanternfly populations," Mar 2025, accessed: 2026-02-27. [Online]. Available: https://a-z-animals.com/articles/how-to-build-a-spotted-lanternfly-trap/

[27] Z. S. Ladin, D. A. Eggen, T. L. E. Trammell, and V. D'Amico, "Human-mediated dispersal drives the spread of the spotted lanternfly (lycorma delicatula)," *Scientific Reports*, vol. 13, Jan 2023, accessed: 2025-05-20. [Online]. Available: https://www.nature.com/articles/s41598-022-25989-3

[28] E. J. E., W. S., T. A., and L. T., "Experimental evidence supports the ability of spotted lanternfly to hitchhike on vehicle exteriors as a mechanism for anthropogenic

dispersal," *Royal Society Open Science*, vol. 11, no. 7, Jul 2024, accessed: 2026-02-18. [Online]. Available: https://royalsocietypublishing.org/rsos/article/11/7/240493/92763/Experimental-evidence-supports-the-ability-of

[29] A. Klappenecker, "Markov chains," 2018, accessed: 2026-02-27. [Online]. Available: https://people.engr.tamu.edu/andreas-klappenecker/csce658-s18/markov_chains.pdf

[30] M. Awiszus and B. Rosenhahn, "Markov chain neural networks," *Workshops of the Computer Vision Foundation*, 2018, accessed: 2026-02-27. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2018_workshops/papers/w42/Awiszus_Markov_Chain_Neural_CVPR_2018_paper.pdf

[31] N. Elias, "Deep learning methodology for early detection and outbreak prediction of invasive species growth," *Proceedings From the IEEE/CVF Winter Conference on Applications of Computer Vision*, vol. 2023, pp. 6335–6343, 2023. [Online]. Available: https://openaccess.thecvf.com/content/WACV2023/html/Elias_Deep_Learning_Methodology_for_Early_Detection_and_Outbreak_Prediction_of_WACV_2023_paper.html

[32] "Ensemble learning," Dec 2025, accessed: 2026-02-27. [Online]. Available: https://www.geeksforgeeks.org/machine-learning/a-comprehensive-guide-to-ensemble-learning/

[33] "Bootstrap method," Jul 2025, accessed: 2026-01-18. [Online]. Available: https://www.geeksforgeeks.org/maths/bootstrap-method/

[34] C. Hayes, "Spotted lanternflies could cost nys grape industry millions," Jan 2025, accessed: 2025-12-02. [Online]. Available: https://news.cornell.edu/stories/2025/01/spotted-lanternflies-could-cost-nys-grape-industry-millions

[35] iNaturalist, "inaturalist," 2025, accessed: 2025-10-07. [Online]. Available: https://inaturalist.org/

[36] E. Detection and D. M. S. (EDDMapS), "Early detection and distribution mapping system (eddmaps)," 2026, accessed: 2026-02-27. [Online]. Available: https://www.eddmaps.org/

[37] GitHub, "Github," 2025, accessed: 2025-10-07. [Online]. Available: https://github.com/

[38] MIT, "Mit license," Jan 2025. [Online]. Available: https://choosealicense.com/licenses/mit/